

- 9 The decomposition of SOCl_2 is a first-order reaction. After 24 hours at a certain temperature, the concentration of SOCl_2 decreases from $0.016 \text{ mol dm}^{-3}$ to $0.002 \text{ mol dm}^{-3}$.

At this temperature, how long would it take for the concentration of SOCl_2 to decrease from $0.016 \text{ mol dm}^{-3}$ to $0.001 \text{ mol dm}^{-3}$?

- A** 8 hours **B** 25.7 hours **C** 32 hours **D** 48 hours